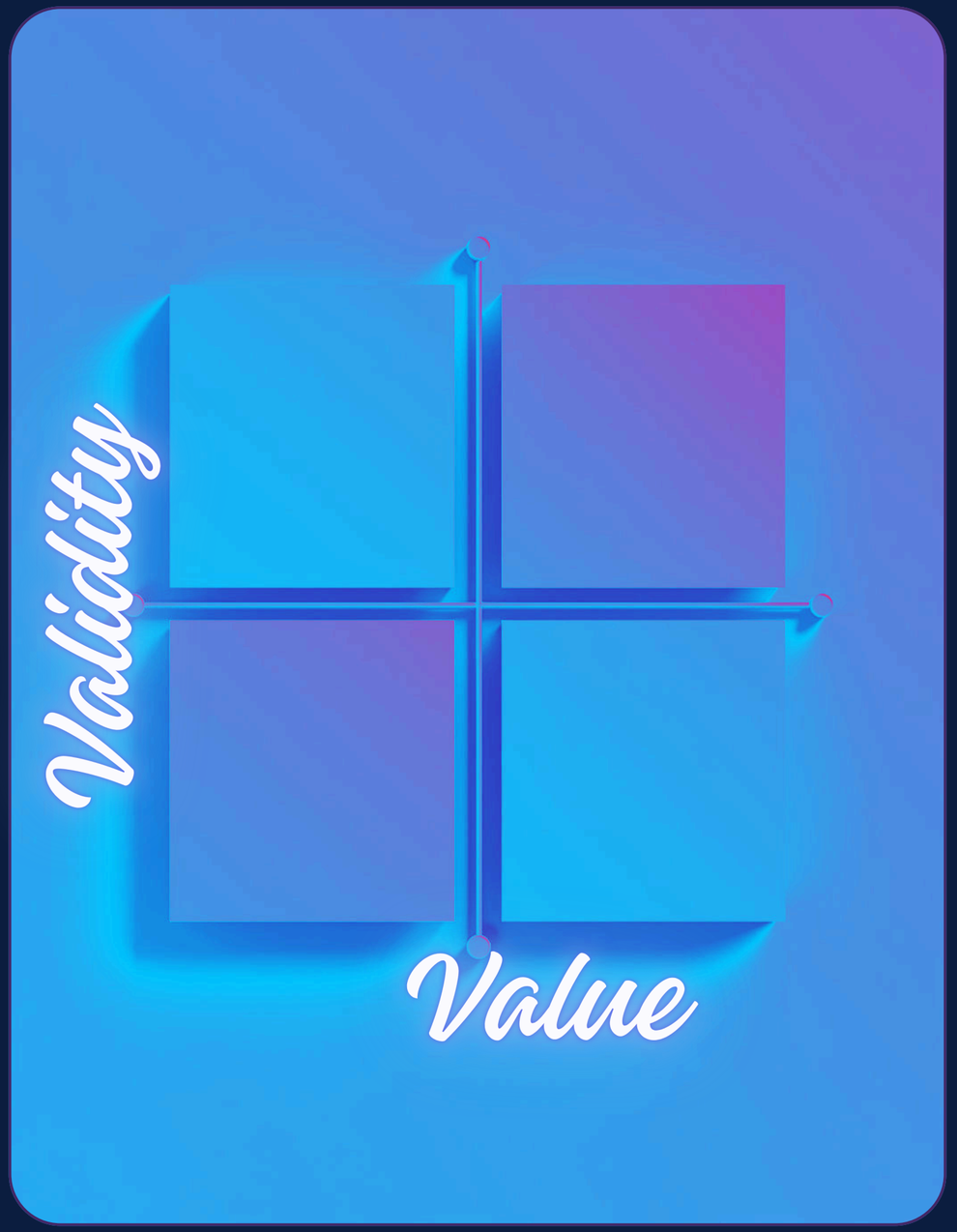


April 5, 2026

VV Matrix Overview

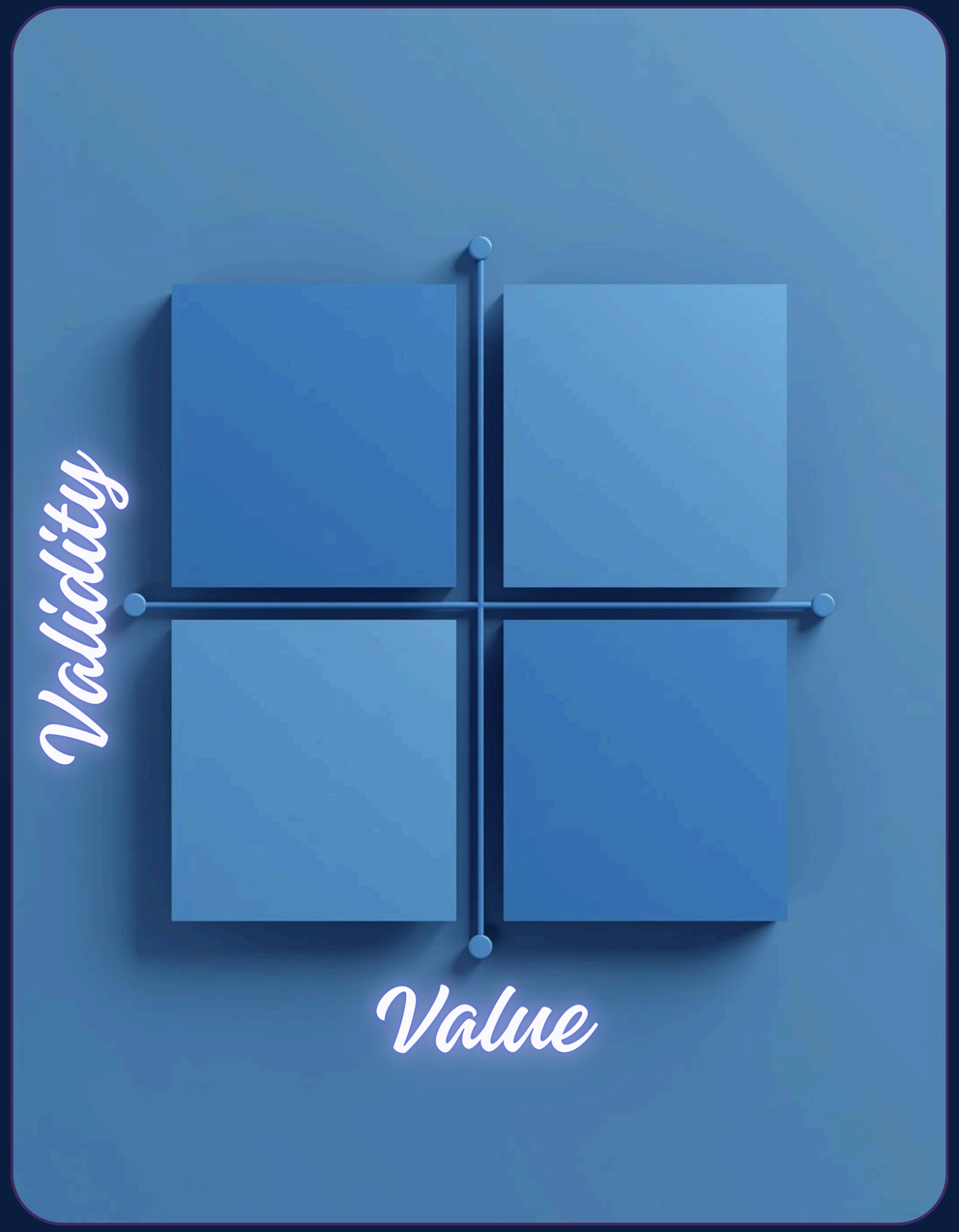
by Weerachon K.

From Stock Report to Risk Management Tool

The VV Matrix

The VV Matrix was developed in April 2026 by Weerachon Khaophong. It was inspired by the need to enhance inventory management efficiency by ensuring **the right stock is maintained at the right level, at the right time.** The objective is to optimize inventory costs while maximizing value creation for the business.

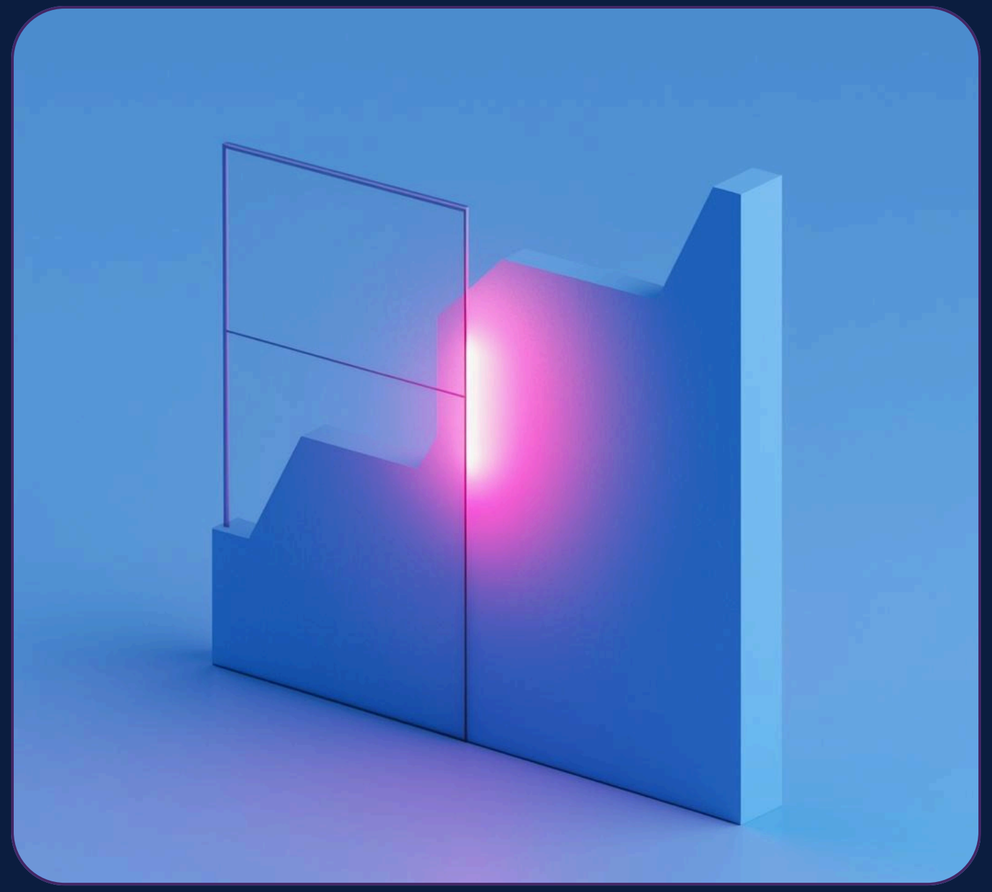
The VV Matrix helps visualize **value and validity**, allowing for better inventory risk management and strategic decision-making.



Traditional Analysis

Traditional inventory analysis relies heavily on aging as a key metric—a lagging indicator that signals issues only after they have materialized.

As a result, decision-making often occurs too late to effectively prevent inventory-related losses.



The exponential VV Matrix goes beyond conventional inventory reporting—it functions as a risk management tool. By applying mathematical weighting, it systematically suppresses the scores of near-expiry products, enabling operations teams to proactively identify risks and execute clearance actions before losses materialize.

Two Dimensions

Understanding Value and Validity Together

Value

Value represents the capital tied up in inventory, indicating how much financial investment is at stake for the organization.

Validity

Validity refers to the remaining shelf life of products, highlighting the urgency to manage inventory effectively before items become obsolete.

Exponential Model

Transforming Risk Assessment in Inventory

New Formula

The formula,
Final Score = Value Score x (Validity Score / 5)$^{\alpha}$, effectively integrates value and freshness into a single impactful metric for inventory management.

Fresh Goods

When products retain high validity scores, they receive maximum value representation, encouraging timely sales and minimizing the risk of obsolescence and financial loss for the organization.

Old Goods

Products approaching expiration dramatically lose their overall score, highlighting the urgency for executives to prioritize clearance efforts and mitigate potential losses before they arise.

Exponential Model

Transforming Risk Assessment in Inventory

New Formula

The formula,

$$\text{Final Score} = \text{Value Score} \times (\text{Validity Score} / 5)^{\alpha}$$

Remark

Variable Definitions

Value Score (V):
A score representing the economic value of the product, typically derived from factors such as purchase value, revenue contribution, or strategic importance.

Validity Score (S):
A score reflecting the product's freshness or remaining shelf life, with a maximum value of 5, where higher scores indicate better validity or longer remaining usability.

α (Alpha):
A weighting parameter that determines the relative importance of freshness (Validity Score) in the overall calculation, allowing the model to adjust the sensitivity to inventory aging through exponential scaling.

Exposing High-Value Products at Risk

Identifying Risks Before They Escalate

This visual emphasizes the need for timely action.

#	Class	Item Code	Item Name	Grp	Stock Value	Expire Date	Days Left	Value	Validity	Simple	Exp Score
50	B A↑	F10100029	Froz. Salmon Trout HeadOn si...	FRM	฿4.7M	2 ส.ค. 2569	118d	5	4	4.4	3.20
51	B A↑	F10100362	Froz. Salmon H/L COHO 10lbs...	FRM	฿4.3M	2 ส.ค. 2569	118d	5	4	4.4	3.20
52	B A↑	F7000100025	Froz. Salmon Gutted 4-5 Kg.(...	FFG	฿4.2M	2 ส.ค. 2569	118d	5	4	4.4	3.20
53	B A↑	F10400187	Aus. Froz. OP Ribs Bone/In M...	FRM	฿484.7K	2 ส.ค. 2569	118d	5	4	4.4	3.20
54	B A↑	F10400222	CHILLED Aus GF 100Days Ch...	FRM	฿472.0K	2 ส.ค. 2569	118d	5	4	4.4	3.20
55	B A↑	F10500050	Aus. Lamb Rack Standard 9 R...	FRM	฿464.1K	2 ส.ค. 2569	118d	5	4	4.4	3.20

Strategic Actions

Classify products for effective management

Class A

Class A items are **high-value** products that remain fresh. They should be readily available and replenished consistently to meet demand and maximize sales opportunities.

Class B

Class B products hold **moderate value** with acceptable validity. Management should follow standard cycles and allocate appropriate storage to ensure these goods remain accessible and organized.

Class C

Class C items are at **risk** due to nearing expiry or prolonged storage. Immediate clearance actions such as promotions or halting orders are crucial to minimize financial loss.

Critical Risk Flag

The Critical Risk Flag identifies potential loss exposure, enabling proactive management of inventory before it must be discounted, written down, or results in expiry-related losses.

Based on exponential score (α=2)

A Strategic — ≥ 3.5 — 11%

49 items · ฿54.8M

B Core — 1.5 – 3.49 — 21%

96 items · ฿53.7M

C At Risk — < 1.5 — 68%

312 items · ฿58.6M

Validity Score Reference

Sc.5	> 180 days	×1.00
Sc.4	91 – 180 days	×0.64
Sc.3	61 – 90 days	×0.36
Sc.2	31 – 60 days	×0.16
Sc.1	≤ 30 days / expired	×0.04

Protect Working Capital

Effective management of inventory risks shields capital, ensuring that resources are allocated efficiently and productively.

Executive Takeaway

Transform risks into opportunities